



MASENO UNIVERSITY

UNIVERSITY EXAMINATIONS 2017/2018

**FIRST YEAR FIRST SEMESTER EXAMINATION FOR
THE DEGREE OF BACHELOR OF SCIENCE IN APPLIED
STATISTICS, ACTUARIAL SCIENCE, MATHEMATICAL
SCIENCE AND MATHEMATICS AND ECONOMICS WITH
INFORMATION TECHNOLOGY**

MAIN CAMPUS

MMA 100: BASIC MATHEMATICS

Date: 20th February, 2018

Time: 3.30 - 6.30pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO
- Electronic scientific calculators may be used
- Observe further instructions on the answer booklet



QUESTION TWO

[20 Marks]

- (a) Using set notations, prove that if A and B are subsets of some universal set, then

$$(A - B) \cup (B - A) = A \cup B - A \cap B.$$

[5 Marks]

- (b) (i) In how many ways can 10 people sit at a round table? [2 Marks]

- (ii) In how many ways can a class of 20 children be split into two groups of 8 and 12 respectively if there are two twins in the class who must not be separated? [3 Marks]

- (iii) How many committees of five people can be chosen from 20 men and 12 women if at least four women must be on each committee? [3 Marks]

- (c) Find the inverse of the matrix $\begin{pmatrix} 1 & 1 & 1 \\ 1 & -2 & -3 \\ 2 & 1 & -1 \end{pmatrix}$ and hence solve for x, y, z in

$$\begin{aligned} x + y + z &= 5 \\ x - 2y - 3z &= -1. \\ 2x + y - z &= 3. \end{aligned}$$

[7 Marks]

QUESTION THREE

[20 Marks]

- (a) Let S_n be the sum of the first n terms of an A.P. series.

- (i) Show that S_n is given by

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

where a is the first term and d is the common difference.

[5 Marks]

- (ii) Find the sum of the first 100 terms of $\sum_{r=1}^n \ln 2^r$ and the least value of n for which the sum of the first $2n$ terms exceeds 1000. [7 Marks]

- (b) Expand $\frac{10}{(1+3x)(1-4x)}$ as a series of ascending powers of x giving the first 4 terms and the range for which the expansion is valid. [8 Marks]



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**FIRST YEAR FIRST SEMESTER EXAMINATION FOR
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INFORMATION TECHNOLOGY**

CITY CAMPUS

MMA 100: BASIC MATHEMATICS

Date: 28th April, 2018

Time: 8.30 - 11.30am

INSTRUCTIONS:

- Answer Question ONE and any other TWO
- Electronic scientific calculators may be used
- Cheating is NOT allowed and will be harshly punished
- Observe further instructions on the answer booklet



QUESTION TWO

[20 Marks]

- (a) Using set notations, prove that if A and B are subsets of some universal set, then

$$(A - B) \cup (B - A) = A \cup B - A \cap B.$$

[6 Marks]

- (b) Each day of a 5-day week, a boy takes an apple or an orange or a banana.

[6 Marks]

- (i) In how many different ways can he organize a week's fruit eating?
- (ii) What would be the answer if he were not allowed to take the same fruit everyday?
- (iii) What is the answer if he may not take the same fruits on any two consecutive days?

- (c) Find the inverse of the matrix $\begin{pmatrix} 1 & 1 & 2 \\ -1 & 2 & 1 \\ 3 & -1 & 3 \end{pmatrix}$ and hence solve for x, y, z in

$$\begin{aligned} x + y + 2z &= 3 \\ -x + 2y + z &= 9 \\ 3x - y + 3z &= 0. \end{aligned}$$

[8 Marks]

QUESTION THREE

[20 Marks]

- (a) Let S_n be the sum of the first n terms of an A.P. series.

- (i) Show that S_n is given by

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

where a is the first term and d is the common difference.

[5 Marks]

- (ii) Find the sum of the first 10 terms of $\sum_{r=1}^n \ln 2^r$ and the least value of n for which the sum of the first $2n$ terms exceeds 1000.

[7 Marks]

- (b) Expand $\frac{10}{(1+3x)(1-2x)}$ as a series of ascending powers of x giving the first 4 terms and the range for which the expansion is valid.

[8 Marks]



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**FIRST YEAR SECOND SEMESTER EXAMINATIONS FOR THE
DEGREE OF BACHELOR OF EDUCATION, BACHELOR OF
SCIENCE AND BACHELOR OF ARTS WITH INFORMATION
TECHNOLOGY**

CITY CAMPUS

MMA 101: ANALYTICAL GEOMETRY

Date: 1st October, 2017

Time: 8.30 - 11.30 am

INSTRUCTIONS:

- Answer question ONE and any other TWO questions
- Electronic scientific calculators may be used
- Observe further instructions on the answer booklet



QUESTION TWO

[20 Marks]

- (a) Given two parallel lines $L_1 : 4y - 3x = -3$ and $L_2 : 4y = 3x - 28$. Determine the distance between the lines L_1 and L_2 and the equation of a line that passes midway between them. [6 Marks]
- (b) Find the equation of a circle which passes through the points $(6, -5)$, $(2, -7)$ and $(-6, -1)$ giving your answer in the form $Ax^2 + By^2 + Cx + Dy + E = 0$. Determine the area of the circle. [10 Marks]
- (c) Give the cartesian coordinates of the following polar coordinates: [4 Marks]
- (i) $(5, 70^\circ)$
 - (ii) $(-3, -300^\circ)$.

QUESTION THREE

[20 Marks]

The equation of an ellipse on the xy -plane is given by $9x^2 + 4y^2 + 36x - 8y + 4 = 0$. Find

- (a) the centre of the ellipse [5 Marks]
- (b) the coordinates of the vertices [4 Marks]
- (c) the foci [3 Marks]
- (d) the eccentricity [2 Marks]
- (e) the directrices [3 Marks]
- (f) the area of the ellipse [3 Marks]

QUESTION FOUR

[20 Marks]

- (a) Consider the equation of a conic section given by $9y^2 - 16x^2 - 144 = 0$.
- (i) Show that the equation represent a hyperbola. [2 Marks]
 - (ii) Determine
 - * the centre [1 Mark]
 - * the foci [3 Marks]
 - * the vertices [2 Marks]
 - * the directrices [2 Marks]
 - * the eccentricity [2 Marks]
 - * the asymptotes. [4 Marks]
- (b) Find the parametric representation of the following cartesian equations: [4 Marks]
- (i) $2x - 1 = 3y$
 - (ii) $x^2 + y^2 = 4$.



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MAIN CAMPUS

MMA 101: ANALYTIC GEOMETRY

Date: 23rd February, 2018

Time: 8.30 - 11.30am

INSTRUCTIONS:

- Answer Question ONE and any other TWO
- Scientific calculators may be used
- Observe further instructions on the answer booklet



Question 2 [20 Marks]

(a) Given

$$x = x' \cos \theta - y' \sin \theta \quad \text{and} \quad y = x' \sin \theta + y' \cos \theta$$

Derive the angle through which axes can be rotated. [10 Marks]

(b) Using rotation of axes formula, analyze the equation given below

$$3x^2 - 2\sqrt{3}xy + y^2 + 2x + 2\sqrt{3}y = 0$$

[10 Marks]

Question 3 [20 Marks]

(a) From 1995 through 2014, the estimated number I (in millions) of adults in the United States who use internet can be modeled by the equation

$$I = 0.025t^2 + 16.9t - 50 \quad 5 \leq t \leq 24$$

where t represents time in years with $t=5$ corresponding to the year 1995. According to the model, in which year did the number of US adults who use the internet hit the 94 million mark? [5 Marks]

(b) A conic section is given by the equation

$$4x^2 + y^2 + 24x - 2y + 21 = 0$$

Find the center, vertices and the foci. [7 Marks]

(c)

i) Determine collinearity of the points $(-2, -2)$, $(1, 1)$ and $(7, 5)$ [5 Marks]

ii) Use the discriminant method to identify the conic given by the equation

$$3x^2 + 5xy - 2y^2 + x - y + 4 = 0$$

[3 Marks]



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ARTS AND BACHELOR OF EDUCATION WITH
INFORMATION TECHNOLOGY**

CITY CAMPUS

MMA 101: ANALYTICAL GEOMETRY

Date: 27th April 2018

Time 2.30 - 5.30pm

INSTRUCTIONS:

- Answer Question ONE (30Marks) and any other TWO questions (20 Marks each).
- Electronic Scientific calculators may be used.
- Observe further instructions in the answer booklet.



QUESTION TWO

[20 Marks]

- (a) Given two parallel lines $L_1 : 4y - 3x = -3$ and $L_2 : 4y = 3x - 28$. Determine the distance between the lines L_1 and L_2 and the equation of a line that passes midway between them. [6 Marks]
- (b) Find the equation of a circle which passes through the points $(6, -5)$, $(2, -7)$ and $(-6, -1)$ giving your answer in the form $Ax^2 + By^2 + Cx + Dy + E = 0$. Determine the area of the circle. [10 Marks]
- (c) Give the cartesian coordinates of the following polar coordinates: [4 Marks]
- (i) $(5, 70^\circ)$
- (ii) $(-3, -300^\circ)$.

QUESTION THREE

[20 Marks]

The equation of an ellipse on the xy -plane is given by $9x^2 + 4y^2 + 36x - 8y + 4 = 0$. Find

- (a) the centre of the ellipse [5 Marks]
- (b) the coordinates of the vertices [4 Marks]
- (c) the foci [3 Marks]
- (d) the eccentricity [2 Marks]
- (e) the directrices [3 Marks]
- (f) the area of the ellipse [3 Marks]

QUESTION FOUR

[20 Marks]

- (a) Consider the equation of a conic section given by $9y^2 - 16x^2 - 144 = 0$.
- (i) Show that the equation represent a hyperbola. [2 Marks]
- (ii) Determine
- * the centre [1 Mark]
 - * the foci [3 Marks]
 - * the vertices [2 Marks]
 - * the directrices [2 Marks]
 - * the eccentricity [2 Marks]
 - * the asymptotes. [4 Marks]
- (b) Find the parametric representation of the following cartesian equations: [4 Marks]
- (i) $2x - 1 = 3y$
- (ii) $x^2 + y^2 = 4$.



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TECHNOLOGY**

CITY CAMPUS

MMA 102: CALCULUS I

Date: 30th September, 2017

Time: 3.30 - 6.30 pm

INSTRUCTIONS:

- Answer question ONE and any other TWO questions
- Electronic scientific calculators may be used
- Observe further instructions on the answer booklet



QUESTION TWO

[20 Marks]

Find $\frac{dy}{dx}$ in the following functions

(a) $y^2 = \cos(\sin \sqrt{x^2 + 1})$ [4 mks]

(b) $x^2 = \frac{x-y}{x+y}$ [3 mks]

(c) $y = \cos^{-1} 2x$ [3 mks]

(d) $y = x^3 \ln 2x$ [3 mks]

(e) $y = x^{x^x}$ [4 mks]

(f) $y = \frac{(x^2+9)^2(x-3)^4}{x^2+2}$ [3 mks]

QUESTION THREE

[20 Marks]

(a) By differentiating $x^2 - y^2 = 1$ implicitly, show that

$$y'' = -\frac{1}{y^3}$$

[5 mks]

(b) Write the polynomial

$$f(x) = -35 + 45x - 20x^2 + 3x^3$$

in the form

$$f(x) = C_0 + C_1(x-3) + C_2(x-3)^2 + C_3(x-3)^3$$

where the C_i 's ($i = 0, 1, 2, 3$) are constants.

[6 mks]

(c) Consider the curve $y = x^5 - 5x^3$

(i) Classify all the stationary points of the curve, and hence

[5 mks]

(ii) sketch the curve

[4 mks]



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MAIN CAMPUS

MMA 104/MAS 101: DESCRIPTIVE STATISTICS

Date: 21st February, 2018

Time: 12.00 - 3.00pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO.



iv) Describe the skewness of the distribution and suggest a reason for it. (3 marks)

QUESTION 2 (20 Marks)

- a) The three measures of central tendency are the mode, median and mean. State when each measure of tendency would be used to describe data (9Marks)
- b) A soccer fan collected data on the number of minutes of league football, m , played by each team in the four main divisions before first scoring a goal at the start of a new season. Her results are shown in the table below.

m (minutes)	Number of teams
$0 \leq m < 40$	36
$40 \leq m < 80$	28
$80 \leq m < 120$	10
$120 \leq m < 160$	4
$160 \leq m < 200$	5
$200 \leq m < 300$	4
$300 \leq m < 400$	2
$400 \leq m < 600$	3

- i) Calculate estimates of the mean and standard deviation of these data. (8 marks)
- ii) Explain why the mean and standard deviation might not be the best summary statistics to use with these data. (2 marks)
- iii) Suggest alternative summary statistics that would better represent these data. (1 mark)

QUESTION 3 (20 Marks)

The number of patients attending a hospital trauma clinic each day was recorded over several months, giving the data in the table below.

Number of patients	10 - 19	20 - 29	30 - 34	35 - 39	40 - 44	45 - 49	50 - 69
Frequency	2	18	24	30	27	14	5

iii) What percent of these respondents are very likely to buy the calendar?

(1 Marks)

iv) What percent of the respondents who are very likely to buy the calendar are alumni?

(1 Marks)

v) Of the alumni, what percent are very likely to buy the calendar?

(1 Marks)

vi) What is the marginal distribution of the campus constituents?

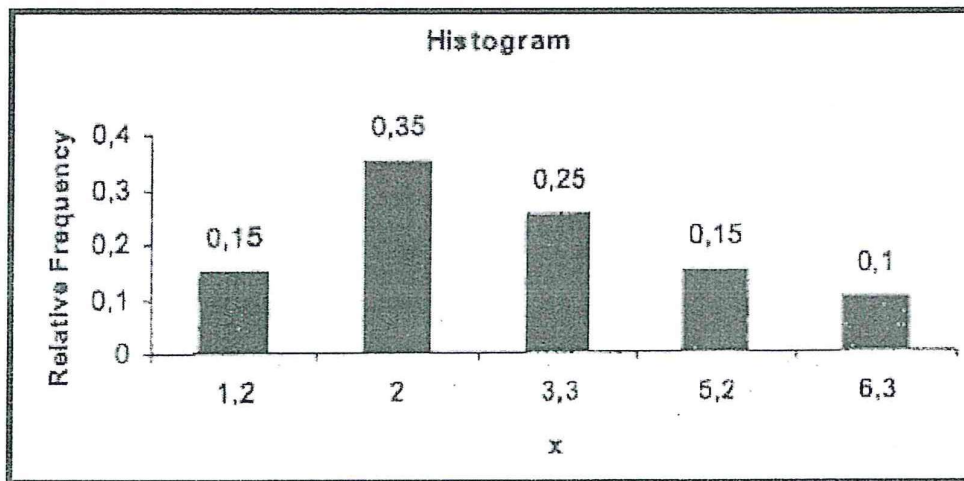
(2 Marks)

vii) What is the conditional distribution of the campus constituents among those very likely to buy the calendar?

(2 Marks)

QUESTION 5 (20 Marks)

The histogram of a discrete data set consisting of 100 measurements is given below:



Determine

a) the mode, median and mean

(5 Marks)

b) the inter-quartile range

(5 Marks)

c) the frequency of 2

(2 Marks)

d) the sum of the data points

(3 Marks)

e) the value of $\sum_{i=1}^N x_i^2$ given that the variance is 2.687

(5 Marks)



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MAIN CAMPUS

MMA 106: MATHEMATICS I

Date: 20th February, 2018

Time: 12.00 - 3.00pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO.



Question 3 (20 marks)

- a) Plot the graph of $y = -2x^2 + 3x + 6$ for values of x from $x = -2$ to $x = 4$. Use the graph to find the roots of the following equations:

i) $-2x^2 + 3x + 6 = 0$

ii) $-2x^2 + 3x + 2 = 0$

iii) $-2x^2 + 3x + 9 = 0$

iv) $-2x^2 + x + 5 = 0$

(17mks)

- b) Using the laws of indices, simplify $\frac{(\frac{4}{3})^3 \times (\frac{3}{5})^{-2}}{(\frac{2}{5})^{-3}}$.

(3mks)

Question 4 (20 marks)

- a) A mobile telephone service provider is undertaking a promotion in which 40% of the scratch cards reveal a prize. A customer has in possession three of the promotional scratch cards. Required:

i) A tree diagram to represent the customers probability space

ii) The probability of receiving one prize

iii) The probability of receiving no prize

iv) The probability of receiving two prizes

v) The probability of winning a prize

(10mks)

- b) i) Prove that $\sin 3A = 3 \sin A - 4 \sin^3 A$.

(6mks)

ii) Express $\frac{1 + \tan 60^\circ}{1 - \tan 60^\circ}$ in surd form and rationalize the denominator.

(4mks)



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**FIRST YEAR FIRST SEMESTER EXAMINATIONS FOR THE
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MAIN CAMPUS

MMA 111: INTRODUCTION TO CALCULUS

Date: 12th February, 2018

Time: 3.30 - 6.30 pm

INSTRUCTIONS:

- Answer question ONE and any other TWO questions
- Start each question on a fresh page
- Indicate question numbers clearly at the top of each page
- Observe further instructions on the answer booklet



b) Find the gradient of the curve

$$x = \frac{t}{1+t}, y = \frac{t^3}{1+t}$$

at the point $(\frac{1}{2}, \frac{1}{2})$ [8mks]

c) Use the Intermediate Value Theorem to show that

$$P(x) = 2x^3 - 5x^2 - 10x + 5$$

has a root somewhere in the interval $[-1, 2]$ [6mks]

QUESTION THREE [20marks]

a) Find the derivative of the following functions

i) $y = \frac{\sec x}{1+\tan x}$ [3mks]

ii) $y = \left(\frac{t-2}{2t+1}\right)^9$ [3mks]

iii) $x^3 + y^3 = 6xy$ [3mks]

iv) $y = \tan^{-1}(e^{2x})$ [3mks]

b) Evaluate the following limits if they exist.

i) $\lim_{x \rightarrow -1} \frac{t+1}{|t+1|}$ [2mks]

ii) $\lim_{x \rightarrow 0} \frac{\cos x + 2x - 1}{3x}$ [2mks]

c) A population of 500 bacteria is introduced into a culture and grows in a number according to the equation

$$P(t) = 500 \left(1 + \frac{4t}{50 + t^2} \right)$$

where t is measured in hours. Find the rate at which the population is growing when $t = 2$. [4mks]

QUESTION FOUR [20marks]

a) Two posts, one 12 feet high and the other 28 feet high, stand 30 feet apart. They are to be stayed by two wires, attached to a single stake, running from ground level to the top of each post. Where should the stake be placed to use the least wire? [10mks]

b) A football team plays in a stadium that holds 55,000 spectators. With ticket prices at ksh.1000, the average attendance had been 27,000. When ticket prices were lowered to ksh.800, the average attendance rose to 33,000.

i) Find the demand(price) function, assuming that it is linear. [5mks]

ii) How should ticket prices be set to maximize revenue? [5mks]



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MAIN CAMPUS

MMA 200: CALCULUS II

Date: 21st February, 2018

Time: 3.30 - 6.30pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO.



Question 3 (20 marks)

- a) State the Fundamental Theorem of Calculus and give its proof. (8mks)
- b) Calculate the area bounded by the curve $y = 3 - x^2$ and the line $y = -1$
- i) by integrating with respect to x and
- ii) by integrating with respect to y . (8mks)
- c) Find the mean value of $i(t) = 20 + 2\sin(\pi t)$ for $t = 5$ to $t = 0.5$. (4mks)

Question 4 (20 marks)

- a) Using the disk method show that the volume, V , of a sphere of radius r is given by the formula

$$V = \frac{4}{3}\pi r^3$$

(6mks)

- b) Use the trapezoidal rule to approximate

$$\int_0^{\pi} \sin x dx$$

for $n = 4$ and $n = 8$ and by finding the exact area compare the results. (10mks)

- c) Evaluate $\int \sin^5 x \cos^3 x dx$. (4mks)

Question 5 (20 marks)

- a) A ball is thrown vertically upwards with an initial velocity of 60 feet per second. How high will the ball go. (Take $a(t) = -32 ft/sec^2$ as the acceleration due to gravity). (10mks)
- b) Calculate the length traced by a particle $p(x, y)$ from time $t = 0$ to $t = \frac{\pi}{2}$ when $x = \sin^2 t$ and $y = \cos^2 t$ (6mks)
- c) Find $\int_1^3 x^2 dx$ using Simpson's Rule with $h = 0.5$ (4mks)



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**SECOND YEAR SECOND SEMESTER EXAMINATION
FOR THE DEGREE OF BACHELOR OF EDUCATION WITH
INFORMATION TECHNOLOGY**

CITY CAMPUS

MMA 201: LINEAR ALGEBRA II

Date: 29th April, 2018

Time: 8.30 - 11.30am

INSTRUCTIONS:

- Answer Question ONE and any other TWO
- Proofs should be written carefully
- Cheating is NOT allowed and will be harshly punished
- Observe further instructions on the answer booklet



QUESTION TWO

[20 Marks]

- (a) Consider the linear space V that consists of all functions of the form $f(t) = c_1 \cos t + c_2 \sin t$ where c_1 and c_2 are arbitrary constants.
- (i) Find the matrix of the linear transformation $T : V \rightarrow V$ given by $T(f) = f'' + af' + bf$ with respect to the basis $\{\cos t, \sin t\}$ where a and b are arbitrary constants. [5 mks]
- (ii) Give a condition for the isomorphism of T . [3 mks]
- (iii) Find the functions f in V such that [5 mks]

$$T(f) = f'' + af' + bf = \cos t.$$

- (b) Evaluate the determinant of the following matrix [7 mks]

$$\begin{pmatrix} -5 & 1 & 1 & -6 & 2 \\ 0 & 1 & 3 & 8 & -4 \\ -1 & 2 & 1 & 8 & 9 \\ 0 & 0 & 1 & 14 & 2 \\ 1 & 1 & 0 & 0 & 0 \end{pmatrix}$$

QUESTION THREE

[20 Marks]

- (a) Determine whether or not the matrix

$$H = \begin{bmatrix} 15 & 8i & 6 - 2i \\ -8i & 0 & -4 + i \\ 6 + 2i & -4 - i & -3 \end{bmatrix}$$

is Hermitian. [5 mks]

- (b) Let $Z = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix}$ be any complex $n \times 1$ matrix. Show that if S is skew-hermitian, then $\bar{Z}^T S Z$ is either zero or pure imaginary. [5 mks]

- (c) Let S denote the space of all skew-symmetric 4×4 matrices.

- (i) Find the basis of S . [7 mks]
- (ii) Determine the dimension of S . [3 mks]



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**SECOND YEAR SECOND SEMESTER EXAMINATIONS FOR
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CITY CAMPUS

MMA 202: VECTOR ANALYSIS

Date: 30th September, 2017

Time: 12.00 - 3.00 pm

INSTRUCTIONS:

- Answer question ONE and any other TWO questions
- Proofs should be written carefully
- Cheating is NOT allowed and will be harshly punished
- Observe further instructions on the answer booklet



QUESTION THREE

[20 Marks]

- (a) If $\mathbf{A} = (3x^2 + 6y)\mathbf{i} - 14yz\mathbf{j} + 20xz^2\mathbf{k}$. Evaluate $\int_C \mathbf{A} \cdot d\mathbf{R}$ from $(0, 0, 0)$ to $(1, 1, 1)$ along the following curves C .

(i) $x = t, y = t^2, z = t^3$. [3 Marks]

- (ii) the straight lines from $(0, 0, 0)$ to $(1, 0, 0)$ then to $(1, 1, 0)$ and finally to $(1, 1, 1)$. [4 Marks]

- (iii) the straight line joining $(0, 0, 0)$ to $(1, 1, 1)$. [3 Marks]

- (b) (i) State Green's theorem. [2 Marks]

- (ii) Evaluate the integral along the curve C .

$$\int_C [(y - x^2 e^x)dx + (\cos 2y^2 - x)dy]$$

where C is the rectangle with vertices $(1, 1)$, $(0, 1)$, $(1, 3)$ and $(0, 3)$. [8 Marks]

QUESTION FOUR

[20 Marks]

- (a) Determine the constant c such that the vector $\mathbf{V} = (x + 3y)\mathbf{i} + (y - 2z)\mathbf{j} + (x + cz)\mathbf{k}$ is solenoidal. [5 Marks]

- (b) (i) State Stoke's theorem. [2 Marks]

- (ii) Verify Stoke's theorem for $\mathbf{A} = (2x - y)\mathbf{i} - yz^2\mathbf{j} - y^2zk$ where S is the upper half surface of the sphere $x^2 + y^2 + z^2 = 1$ and C is its boundary. [13 Marks]

QUESTION FIVE

[20 Marks]

- (a) Evaluate

$$\iint_S \mathbf{A} \cdot \mathbf{n} dS$$

where $\mathbf{A} = 18z\mathbf{i} - 12\mathbf{j} + 3y\mathbf{k}$ and S is that part of the plane $2x + 3y + 6z = 12$ which is located in the first octant. [10 Marks]

- (b) (i) State divergence theorem. [2 Marks]

- (ii) Use divergence theorem to evaluate

$$\iint_S \mathbf{A} \cdot \mathbf{n} dS$$

where $\mathbf{A} = 4xz\mathbf{i} - y^2\mathbf{j} + yz\mathbf{k}$ and S is the surface of the cube bounded by $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$. [8 Marks]



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WITH INFORMATION TECHNOLOGY**

MAIN CAMPUS

MMA 202: VECTOR ANALYSIS

Date: 13th March, 2018

Time: 3.30 - 6.30pm

INSTRUCTIONS:

- Attempt Question ONE and any other TWO.
- Observe further instructions from the examination answer book



Question 3

- Define the term 'line integral' of a continuous vector field (3mks)
- Find the work done by $\vec{F} = (y - x^2)i + (z - y^2)j + (x - z^2)k$ over the curve $r(t) = ti + t^2j + t^3k$ from $(0, 0, 0)$ to $(1, 1, 1)$ (7mks)
- Let $\vec{F} = x^2 + y^2 + z^2$. Evaluate $\iiint_v F dv$ where v is the region bounded by $x + y + z = a$ ($a > 0$), $x = 0$, $y = 0$, $z = 0$ (10mks)

Question 4

- Evaluate $\int_c \vec{A} \cdot d\vec{r}$ where $\vec{A} = 3xyi - y^2j$ along $y = 2x^2$ in the $x - y$ plane from $(0, 0)$ to $(1, 2)$ (5mks)
- State the divergence theorem hence evaluate $\iiint_s \vec{F} \cdot \vec{n} ds$ where the vector $\vec{F} = 4xz i - y^2 j + yz k$ and s is the surface of a cube bounded by $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$ (10mks)
- State and prove the Frenet Serret formulas (5mks)

Question 5

- Verify Greens theorem for $\oint_c (xy + y^2)dx + x^2 dy$ where c is the region bounded by the curves $y = x$ and $y = x^2$ (8mks)
- Verify the Stokes' theorem for the vector field $\vec{F} = 2zi + 3xj + 5yk$ taking s to be the portion of the paraboloid $z = 4 - x^2 - y^2$ for which $z \geq 0$ with upward orientation and the curve to be positively oriented circle $x^2 + y^2 = 4$ that forms the boundary of s in the xy plane (12mks)

END
ALL THE BEST



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**SECOND YEAR FIRST SEMESTER EXAMINATIONS FOR THE
DEGREE OF BACHELOR OF EDUCATION SCIENCE WITH
INFORMATION TECHNOLOGY**

CITY CAMPUS

**MMA 210: PROBABILITY AND DISTRIBUTION
THEORY I**

Date: 29th September, 2017

Time: 2.30 - 5.30 pm

INSTRUCTIONS:

- Answer question ONE and any other TWO questions
- Show all the necessary workings
- Start each question on a new page



QUESTION TWO (20mks)

- (a) The distribution function for a random variable x is

$$F(x) = \begin{cases} 1 - e^{-2x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Find;

- (i) the density function [2 mk]
- (ii) the probability that $x > 2$ [3mks]
- (iii) the probability that $-3 < x \leq 4$ [3mks]

- (b) If the random variable x and y have joint density function

$$f(x, y) = \begin{cases} \frac{xy}{96}, & 0 < x < 4, 1 < y < 5 \\ 0, & \text{otherwise} \end{cases}$$

Find the density function of $u = x + 2y$ [5mks]

- (c) For a Gamma distribution with parameters α and β , show that $\mu = \alpha\beta$, given that $\Gamma(\alpha) = \int_0^\infty y^{\alpha-1} e^{-y} dy$ and that $y = \frac{x}{\beta}$ [7mks]

QUESTION THREE (20mks)

- (a) Suppose a random variable x has the exponential distribution

$$f(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

- (i) Find the m.g.f of the distribution [2mks]
 - (ii) Find $E(x)$ and $\text{Var}(x)$ [8mks]
- (b) Let (x, y) be a two dimensional random variable having the joint pdf

$$f(x, y) = \begin{cases} e^{-(x+y)}; & 0 < x < \infty, 0 < y < \infty \\ 0; & \text{elsewhere} \end{cases}$$

Let $u = x + y$ and $v = \frac{x}{x+y}$, find



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BACHELOR OF EDUCATION WITH INFORMATION
TECHNOLOGY**

MAIN CAMPUS

MMA 210: PROBABILITY DISTRIBUTION THEORY I

Date: 15th February, 2018

Time: 3.30 - 6.30 pm

INSTRUCTIONS:

- Answer question ONE and any other TWO questions



iv).determine the joint sample space (X,Y) [3mks]

v).determine the joint probability distribution of X and Y [4mks]

vi).from the joint probability distribution $f(x,y)$ obtain the distribution of Y [4mks]

QUESTION THREE (20MKS)

a). Suppose that X and Y are continuous random variables such that (X,Y) takes the values only in the rectangle $\{(x,y): 0 \leq x \leq 2, 0 \leq y \leq 1\}$. Suppose also that the joint distribution function of X and Y at any point (x,y) in the rectangle is given by

$$F(x,y) = \frac{1}{90}xy(x^2 + y^2)$$

i) determine $P(1 \leq x \leq 3, \frac{1}{2} \leq y \leq 1)$ [4mks]

ii) determine the joint p.d.f of X and Y [4mks]

b). Let X and Y be independent random variables each having an exponential distribution with parameter λ . Find the distribution of X+Y [6mks]

c). Let X and Y be independent random variables each having a geometric distribution with parameter p. Find the distribution of X+Y [6mks]

QUESTION FOUR (20MKS)

a). Suppose that $f(x,y) = x + y$ $0 < x < 1, 0 < y < 1$

i). compute $E(XY)$ [3mks]

ii). compute $E(3X + 5Y)$ [3mks]

b). Let X and Y have the joint p.d.f $f(x,y) = \begin{cases} x + y & 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$

i). Determine the conditional mean of Y given $X = x, 0 < x < 1$ [9mks]

ii). Determine the conditional variance of Y given $X = x, 0 < x < 1$ [5mks]

QUESTION FIVE (20MKS)

a) Find the joint p.d.f of the two random variables X and Y whose joint cumulative function is [5mks]

$$F(X,Y) = \begin{cases} (1 - e^{-x})(1 - e^{-y}) & x > 0, y > 0 \\ 0 & x < 0, y < 0 \end{cases}$$

b). Hence use the joint p.d.f to determine $P(1 < x < 3, 1 < y < 2)$ [5mks]

c). If $f(x,y) = \begin{cases} \frac{2}{3}(x + 2y) & 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$



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**SECOND YEAR SECOND SEMESTER EXAMINATIONS FOR
THE DEGREE OF BACHELOR OF EDUCATION SCIENCE
WITH INFORMATION TECHNOLOGY**

CITY CAMPUS

**MMA 211: PROBABILITY AND DISTRIBUTION
THEORY II**

Date: 1st October, 2017

Time: 8.30 - 11.30 am

INSTRUCTIONS:

- Answer question ONE and any other TWO questions
- Show all the necessary workings
- Begin each question on a new page



(e) Let a discrete population be given by the numbers $\{1, 2, 3\}$. If all random samples of size 2 are drawn without replacement, obtain the following in each case

(i) The sampling distribution of the sample mean

(ii) The mean and variance of the sample mean. [6mks]

QUESTION TWO (20mks)

(a) (i) State and prove Chebyshev's Inequality

(ii) A random variable X has mean 24 and variance 9. Obtain a bound on the probability that the random variable X assumes values between 16.5 to 31.5 [10mks]

(b) Let the random variable x and y have a joint density function

$$f(x, y) = \begin{cases} \frac{1}{2}(x + y)e^{-(x+y)}, & x, y > 0 \\ 0, & \text{elsewhere} \end{cases}$$

Find the density function of $z = x + y$ [5mks]

(c) A fruit drink company wants to know the variation, as measured by the standard deviation, of the amount of juice in 16 ounce cans. From past experience, it is known that $\sigma^2 = 2$. The company statistician decides to take a sample of 25 cans from the population line and compute the sample variance. Assuming that the sample values may be viewed as a random sample from a normal population, find a value of b such that $p(S^2 > b) = 0.05$, (S^2 - sample variance) [5mks]

QUESTION FOUR (20mks)

- (a) A manufacturer of screws has noticed that on average, 0.02 proportion of screws produced are defective. A random sample of 400 screws is examined for the proportion of defective screws. Find the probability that the proportion of the defective screws $\bar{p}$ in the sample is between 0.01 and 0.03? [5mks]
- (b) Sixteen oil tins are taken at random from an automatic filling machine. The mean weight of the tins is 14.2 kg, with a standard deviation of 0.40 kg. Can we conclude that the filling machine is wasting oil by filling more than the intended weight of 14 kg, at a significance level of 5%? [5mks]
- (c) Five hundred ball bearings have a mean weight of 5.02 oz and a standard deviation of 0.30 oz. Find the probability that a random sample of 100 ball bearings chosen from this group will have a combined weight of more than 510 oz. [4mks]
- (d) The mean weekly sale of soap bars in departmental stores was 146.3 bars per store, after an advertisement campaign, the mean weekly sale in 22 stores for a typical week increased to 153.7 and shown a standard deviation of 17.2. Was the advertisement campaign successful at 1% significance level? [6mks]

QUESTION FIVE (20mks)

- (a) (i) Let T follow a t -distribution with $r = 8$ df. What is the probability that the absolute value of T is less than 2.306? [5mks]
- (ii) A company did a survey of 20 people who used its product. The mean average age of the sample was 22.4 years and the unbiased standard deviation was 3.1 years. The company now wants to advertise in a magazine that has a primary readership of people who are aged between 18 and 24. What percent of its potential customers are in this age bracket? [5mks]



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SCIENCE AND MATHEMATICS AND ECONOMICS WITH
INFORMATION TECHNOLOGY**

MAIN CAMPUS

MMA 215: CALCULUS

Date: 23rd February, 2018

Time: 8.30 - 11.30am

INSTRUCTIONS:

- Answer Question ONE and any other TWO



(b) Differentiate the following with respect to x .

(i) $y = \frac{x}{\sqrt{x^2-1}}$

(ii) $y = (\sqrt{x})^x$

[5 Marks]

(c) Integrate the following.

(i) $\int \frac{2x+3}{x^3+4x^2+4x} dx$

(ii) $\int \frac{x+1}{\sqrt{x^2+2x+5}} dx$

[7 Marks]

QUESTION THREE

[20 Marks]

(a) (i) Define uniform continuity of a function f on a closed interval D .

(ii) Show that the function $f(x) = x^2$ is uniformly continuous on the interval $[-1,1]$.

[6 Marks]

(b) (i) Show that $\int e^x [f(x) + f'(x)] dx = e^x f(x)$ and hence, evaluate

(ii) $\int \frac{1+x}{(2+x)^2} e^x dx$.

[9 Marks]

(c) Evaluate $\int_0^{\sqrt[6]{3}} \frac{x^2}{1+x^6} dx$.

[5 Marks]

QUESTION FOUR.

[20 Marks]

(a) (i) Compute the arc length of the graph of $f(x) = x^{3/2}$ over the interval $0 \leq x \leq 1$.

(ii) Determine the surface area of a solid obtained by rotating the curve

$y = \sqrt{9-x^2}$, $-3 \leq x \leq 3$ about the x -axis

[9 Marks]



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SCIENCE AND MATHEMATICS AND ECONOMICS WITH
INFORMATION TECHNOLOGY**

MAIN CAMPUS

MMA 217: NUMERICAL CONVERGENCE

Date: 19th February, 2018

Time: 3.30 - 6.30pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO.
- Scientific calculators may be used
- Observe further instructions from the booklet
- Indicate question numbers clearly at the top of each page



Question 2 [20 Marks]

- (a) Apply Romberg's integration formula to find

$$\int_0^1 \frac{dx}{1+x^2}$$

correct to four-decimal places. Take $h=0.5, 0.25$ and 0.125 [10 Marks]

- (b) Given that

$$\begin{aligned} \sqrt{15500} &= 124.4990 & \sqrt{15510} &= 124.5392 \\ \sqrt{15520} &= 124.5793 & \sqrt{15530} &= 124.6194 \end{aligned}$$

Find the value of $\sqrt{15516}$ [10 Marks]

Question 3 [20 Marks]

- (a) Compute the root of the equation

$$x^3 - 8x - 5 = 0$$

using secant method over the interval $(3,4)$. [10 Marks]

- (b) The following is a table of values of a polynomial degree. If its given that $f(3)$ is in error. Correct the error.

x	0	1	2	3	4	5	6
y = f(x)	1	2	33	254	1054	3126	7777

[10 Marks]

Question 4 [20 Marks]

- (a) Using Lagrange interpolation formula to compute the value of y corresponding to $x=10$ from the following set of data

x	5	6	9	11
y = f(x)	380	-2	196	508

[6 Marks]

- (b) Using Newton's difference formula, evaluate the value of $f(84)$ for the data given in the table below [8 Marks]

x	40	50	60	70	80	90
y = f(x)	204	224	246	270	296	324



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**THIRD YEAR FIRST SEMESTER EXAMINATION FOR
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WITH INFORMATION TECHNOLOGY**

MAIN CAMPUS

MMA 300: REAL ANALYSIS I

Date: 13th March, 2018

Time: 3.30 - 6.30pm

INSTRUCTIONS:

- Attempt Question ONE and any other TWO.
- Show all the necessary workings
- Indicate question numbers clearly at the top of each page



QUESTION TWO [20marks]

a) Prove that the set $\mathbb{R}$ of all real numbers is uncountable. [7mks]

b) Show that the function

$$f(x) = \begin{cases} \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

is continuous but not uniformly continuous on $[0, \infty)$. [8mks]

c) Let (X, d) be any metric space. Show that the function d_1 defined by

$$d_1(x, y) = \frac{d(x, y)}{1 + d(x, y)}$$

is a metric on X . [5mks]

QUESTION THREE [20marks]

a) Prove that if a sequence (x_n) converges then it is bounded. Illustrate with the help of an example that the converse is not necessarily true. [8mks]

b) Show that the sequence (S_n) , where

$$S_n = \frac{1}{1!} + \frac{1}{2!} + \cdots + \frac{1}{n!} \quad \forall n \in \mathbb{N}$$

is convergent. [6mks]

c) Prove that a function f which is uniformly continuous on an interval is continuous on that interval. [6mks]

QUESTION FOUR [20marks]

a) Show that the function $f(x) = 3x^2 + 1$ is Riemann integrable on any interval $[0, 3]$, and that $\int_0^3 f(x)dx = 30$ [8mks]

b) Investigate the convergence or divergence of the series [8mks]

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$



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SCIENCE AND MATHEMATICS AND ECONOMICS WITH
INFORMATION TECHNOLOGY**

MAIN CAMPUS

MMA 301: ORDINARY DIFFERENTIAL EQUATIONS I

Date: 22nd February, 2018

Time: 3.30 - 6.30pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO.
- Scientific calculators may be used
- Observe further instructions from the booklet
- Indicate question numbers clearly at the top of each page



Question 2 [20 Marks]

- (a) A circuit in series has an electromotive force of 40V with a resistance of 10Ω and an inductance of 0.2henrys. If the initial current is zero, solve the basic differential equation to find current i on the circuit at any time $t > 0$ [6 Marks]

- (b) Using variation of parameters, solve the ODE

$$y'' + y = \sec x$$

[8 Marks]

- (c) Find the solution to the ODE given below

$$\frac{dy}{dx} - y = 4y^5$$

[6 Marks]

Question 3 [20 Marks]

- (a) Consider the linear second order differential equation

$$y'' + P_1y' + P_2y = f(x)$$

Derive the formula for particular integral used in solving differential equation by variation of parameters [10 Marks]

- (b) Evaluate the differential equation

$$(1 - x^2)y'' - 2xy' + 2y = 0 \quad y_1 = x$$

[8 Marks]

- (c) Consider the given equation and determine its exactness

$$(2x \sin y + y^3 e^x) + (x^2 \cos y + 3y^2 e^x) \frac{dy}{dx} = 0$$

[2 Marks]



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MAIN CAMPUS

MMA 307: METHODS I

Date: 22nd February, 2018

Time: 8.30 - 11.30am

INSTRUCTIONS:

- Answer Question ONE and any other TWO
- Show all the necessary workings



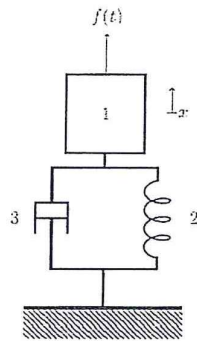


Figure 0.0.1: Mass-spring-damper System

(c) Consider the following system

Suppose at time $t = 0$, this system is at position $x = 0$, but with an initial velocity of 1 m/s and the force $f(t) = e^{-3t}$ is applied. Find $x(t)$ for $t \geq 0$, if the differential equation governing the system is given by

$$\frac{d^2x}{dt^2} + 3\frac{dx}{dt} + 2x = e^{-3t}$$

[8mks]

QUESTION THREE (20mks)

(a) Find the first three terms of the Legendre Series

$$f(x) = \begin{cases} 0, & -1 < x < 0 \\ 1, & 0 \leq x \leq 1 \end{cases}$$

[4mks]

(b) (i) Prove that $\Gamma(\frac{1}{2}) = \sqrt{\pi}$

[6mks]

(ii) Show that the change of dependent variable $y = x^{\frac{1}{2}}z$ transforms

$\frac{d^2y}{dx^2} + y = 0$ into a Bessel equation, hence write the solution of the Bessel's equation in terms of z .

[5mks]



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INFORMATION TECHNOLOGY**

MAIN CAMPUS

MMA 319: NUMERICAL MTHEMATICS I

Date: 20th February, 2018

Time: 8.30 - 11.30am

INSTRUCTIONS:

- Answer Question ONE and any other TWO



QUESTION TWO

- a) Find the approximation to the I.V.P using Euler's method with a step size of 0.1

$$\frac{dy}{dt} = -y + t + 1 \quad ; \quad 0 \leq t \leq 1, \quad y(0) = 1$$

Compare your value with the exact value at $y(0) = 1$ (16 marks)

- b) Show that if the relaxation parameter is unity i.e. $w=1$, then the successive over relaxation (SOR) method reduces to Gauss-Seidel method. (4 marks)

QUESTION THREE

- a) Solve the equation

$$4x_1 + x_2 + 2x_3 = 4$$

$$3x_1 + 5x_2 + x_3 = 7$$

$$x_1 + x_2 + 3x_3 = 3$$

Using Jacobi iterative matrix method starting with initial vector zero and perform **four** iterations. (10 marks)

- b) Use the modified Euler's formula with $h=1$, $h=0.5$ and $h=0.25$ to approximate $y(1)$ given that

$$\frac{dy}{dt} = y + t, \quad y(0) = 1 \quad (10 \text{ Marks})$$

QUESTION FOUR

- a) Determine all the eigenvalues of matrix D. Hence show that one of the eigenvectors corresponding to the eigenvalue, $\lambda = 3$ is $(3, 6)^T$.

$$D = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix} \quad (7 \text{ Marks})$$



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**FOURTH YEAR FIRST SEMESTER EXAMINATIONS FOR
THE DEGREE OF BACHELOR OF EDUCATION WITH
INFORMATION TECHNOLOGY**

CITY CAMPUS

MMA 403: TOPOLOGY I

Date: 29th September, 2017

Time: 8.30 - 11.30 am

INSTRUCTIONS:

- Answer question ONE and any other TWO questions
- Proofs should be written carefully
- Cheating is NOT allowed and will be harshly punished
- Observe further instructions on the answer booklet



QUESTION TWO

[20 Marks]

- (a) Let X be a set and A_i be a family of sets such that $i \in I$ with X and A_i , for all $i \in I$ being considered within the same universal set. Show that

$$X \setminus \bigcap_{i \in I} A_i = \bigcup_{i \in I} X \setminus A_i$$

[5 mks]

- (b) Let $S_k \subseteq \mathbb{R}$ be defined by $S_k = \left(1 - \frac{1}{k}, 2 + \frac{1}{k}\right]$, $k \geq 1$. Find:

(i) $\bigcup_{k=1}^{\infty} S_k^c$

[3 mks]

(ii) $\bigcap_{k=1}^{\infty} S_k^c$

[3 mks]

- (c) Let X be a nonempty set and $\{\tau_\lambda : \lambda \in \Lambda\}$ be any collection of topologies on X .

- (i) Show that $\bigcap_{\lambda \in \Lambda} \tau_\lambda$ is a topology on X .

[6 mks]

- (ii) Give an example to show that the union of two topologies is not necessarily a topology.
[3 mks]

QUESTION THREE

[20 Marks]

- (a) If $(A_i)_{i \in I}$ is an arbitrary collection of subsets of a topological space X , let $\text{int}(A_i)$ denotes the interior of the set A_i .

- (i) Show that $\bigcup_{i \in I} \text{int}(A_i) \subseteq \text{int}(\bigcup_{i \in I} A_i)$.

[4 mks]

- (ii) By using an appropriate example in $\mathbb{R}$, show that the inclusion in (i) above may be proper.
[4 mks]

- (b) Consider the set $X = \{a, b, c, d\}$ and a class of its subsets $\sigma = \{\{a\}, \{b, c\}, \{b, c, d\}, \{a, d, e\}\}$.

- (i) Determine the basis β and the topology τ on X generated by the class σ .

[4 mks]

- (ii) List the closed subsets of X .

[4 mks]

- (iii) Find the closure of the subset $\{b, c, d\}$.

[2 mks]

- (iv) Find the interior of the subset $\{b, c\}$.

[2 mks]



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SCIENCE AND MATHEMATICS AND ECONOMICS WITH
INFORMATION TECHNOLOGY**

MAIN CAMPUS

MMA 403: TOPOLOGY I

Date: 19th February, 2018

Time: 12.00 - 3.00pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO.
- Observe further instructions on the answer booklet
- Proofs should be written carefully



QUESTION FOUR

[20 Marks]

- (a) Let (X, τ) be a topological space and $\beta \subset \tau$. Show that the following two properties of β are equivalent; [10 mks]

(i) β is a basis for τ

(ii) For each $G \in \tau$ and each $x \in G$, there is a $U \in \beta$ with $x \in U \subset G$.

- (b) Let $C[0, 1]$ be the space of continuous functions on $[0, 1]$ and let for all $f \in C[0, 1]$, $\epsilon > 0$ define

$$M(f; \epsilon) = \{g \in C[0, 1] : \sup\{|f(x) - g(x)| : x \in [0, 1]\} < \epsilon\}.$$

Show that $M(f; \epsilon)$ is a basis.

[10 mks]

QUESTION FIVE

[20 Marks]

- (a) Show that any subspace (Y, τ_Y) of a first countable space X, τ is also first countable. [6 mks]

- (b) Define identification topology and then show that identification topology is actually a topology. [8 mks]

- (c) Show that the real line $\mathbb{R}$ is not compact. [6 mks]

END
ALL THE BEST



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**FOURTH YEAR FIRST SEMESTER EXAMINATION FOR
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SCIENCE WITH INFORMATION TECHNOLOGY**

MAIN CAMPUS

MMA 404: COMPLEX ANALYSIS II

Date: 13th March, 2018

Time: 3.30 - 6.30pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO
- Start each question on a fresh page
- Indicate question numbers clearly at the top of each page
- Observe further instructions on the answer book.



Question Four (20 marks)

Determine the integral of

$$f = \frac{-3z + 4}{z(z-1)(z-2)}$$

around the circle $|z| = \frac{3}{2}$

[20 marks]

Question Five: (20marks)

(a) State the Cauchy-Riemann equations for an analytic function $f(z) = u + iv$

[5 marks]

(b) Show that the Cauchy-Riemann equations are satisfied for the equation $f(z) = \ln(z)$

[15 marks]

===== END =====



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**FOURTH YEAR FIRST SEMESTER EXAMINATION FOR
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BACHELOR OF EDUCATION SCIENCE
WITH INFORMATION TECHNOLOGY**

MAIN CAMPUS

MMA 405: PARTIAL DIFFERENTIAL EQUATIONS I

Date: 13th March, 2018

Time: 8.30 - 11.30am

INSTRUCTIONS:

- Attempt Question ONE and any other TWO.
- Show all the necessary workings
- Observe further instructions on the answer booklet



(b) Find the solution of

$$(x^2 - 1)p + xyq + y^2z = x^2 - 1$$

which is zero on the positive y - axis.

[13mks]

[Question Three, 20mks]

(a) Find the characteristics of the equation

$$pq = z$$

and hence, determine the integral surface which passes through the parabola $x = 0, y^2 = z$.

[10mks]

(b) Using charpit's method, find the complete integral of

$$x^2p^2 + y^2q^2 - 1 = 0$$

[10mks]

[Question Four, 20mks]

Apply Jacobi method to find complete integral of the following equations

$$(i) p_1p_2p_3 + p_4^3x_1x_2x_3x_4^3 = 0$$

[10mks]

$$(ii) p_1p_2p_3 = z^3x_1x_2x_3$$

[10mks]

[Question Five, 20mks]

Find the integral surface of the equation

$$x(x+y)p - y(x+y)q - z(x-y) = 0$$

which passes through $x^2 + z^2 = 2x, y = 1$

[20mks]

===== END =====



MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2017/2018

**FOURTH YEAR FIRST SEMESTER EXAMINATION FOR
THE DEGREE OF BACHELOR OF SCIENCE IN APPLIED
STATISTICS, ACTUARIAL SCIENCE, MATHEMATICAL
SCIENCE AND MATHEMATICS AND ECONOMICS WITH
INFORMATION TECHNOLOGY**

MAIN CAMPUS

MMA 409: DIFFERENTIAL GEOMETRY

Date: 23rd February, 2018

Time: 3.30 - 6.30pm

INSTRUCTIONS:

- Answer Question ONE and any other TWO.
- Show all the necessary workings



QUESTION TWO (20mks)

- (a) Show that along the curve $X = X(s)$,

$$X''' = -k^2 \mathbf{t} + k' \mathbf{n} + \tau k \mathbf{b}$$

[4mks]

- (b) Find the principle unit normal and unit binormal along the space curve $\mathbf{r}(t) = (3t - t^3, 3t^2, 3t + t^3)$. Further, prove that the curvature and the torsion along the curve are the same. [16mks]

QUESTION THREE (20mks)

- (a) Find the length of a curve $\vec{r}(t) = t^3 \mathbf{i} + t^2 \mathbf{j}$ from $t = 0$ to $t = 1$ and hence compare it to the straight line distance between the end points of the curve. [8mks]
- (b) Show that the surface $\vec{X} = u\mathbf{e}_1 + v\mathbf{e}_2 + (u^2 + v^3)\mathbf{e}_3$ is elliptic, hyperbolic and parabolic for $v > 0$, $v < 0$ and $v = 0$ respectively. [7mks]
- (c) Show that the surface area of the Monge patch $\mathbf{x} = u\mathbf{e}_1 + v\mathbf{e}_2 + f(uv)\mathbf{e}_3$ is given by

$$A = \int_w \int_u \sqrt{1 + f_u^2 + f_v^2} du dv$$

[5mks]

QUESTION FOUR (20mks)

- (a) Show that the unit tangent vector to the curve

$$\mathbf{r}(t) = (4 \cos t, \cos 2t, 2t + \sin 2t)$$

is $\mathbf{T} = (-\sin t, -\sin t \cos t, \cos^2 t)$ and that the curvature is

$$\frac{1}{4} (1 + \cos^2 t)^{\frac{1}{2}}$$

[9mks]



MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2016/2017

**FOURTH YEAR FIRST SEMESTER EXAMINATIONS FOR
THE DEGREE OF BACHELOR OF SCIENCE WITH
INFORMATION TECHNOLOGY**

CITY CAMPUS

MMA 409: DIFFERENTIAL GEOMETRY

Date: 29th September, 2017

Time: 2.30 - 5.30 pm

INSTRUCTIONS:

- Answer question ONE and any other TWO questions
- Show all the necessary workings
- Start each question on a new page
- Observe further instructions on the answer booklet



QUESTION TWO (20mks)

- (a) Show that the tangent vectors along the curve

$\vec{X} = ate_1 + bt^2e_2 + t^3e_3$, where $2b^2 = 3a$ make a constant angle with the vector $\vec{v} = e_1 + e_3$ [4mks]

- (b) (i) Determine the point at which the curves

$$r_1(\vec{t}) = e^ti + 2 \sin \left[t + \frac{\pi}{2} \right] j + (t^2 - 2)k$$

and

$$r_2(\vec{u}) = ui + 2j + (u^2 - 3)k$$

intersect and find the angle of intersection. [6mks]

- (ii) Find the intersection of the x_1x_2 plane and the tangent line to the helix $\mathbf{x} = \cos t e_1 + \sin t e_2 + t e_3$, ($t > 0$). [4mks]

- (c) Find the curvature along the curve $\mathbf{X} = (1 - \sin t)e_1 + (1 - \cos t)e_2 + e_3$ and hence determine the radius of curvature. [6mks]

QUESTION THREE (20mks)

- (a) Consider the helix $X(t) = (a \cos t)e_1 + (a \sin t)e_2 + bte_3$ where $a > 0$ and $b \neq 0$, show that the curvature is $k(s) = \frac{a}{a^2+b^2}$ and torsion

$$\tau(s) = \frac{b}{\sqrt{(a^2+b^2)}} \quad [15mks]$$

- (b) Show that the first fundamental form on the surface of revolution

$$\vec{X} = (f(t) \cos \theta) e_1 + (f(t) \sin \theta) e_2 + g(t) e_3$$

is

$$I = f^2(t) d\theta^2 + ((f'(t))^2 + (g'(t))^2) dt^2$$

[5mks]